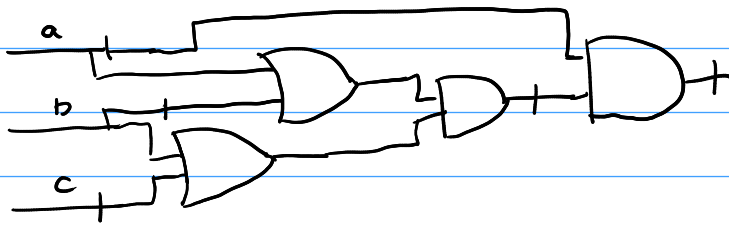
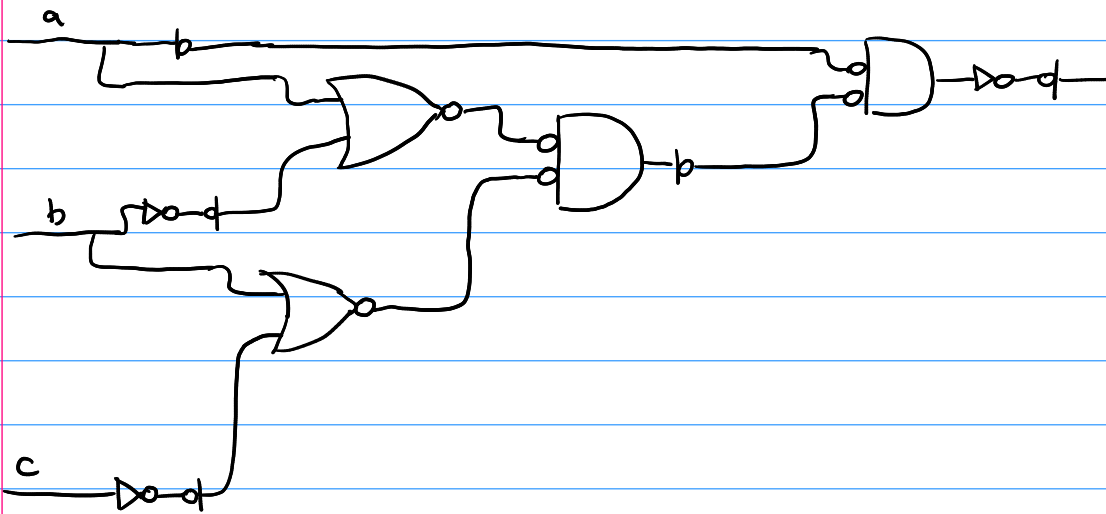


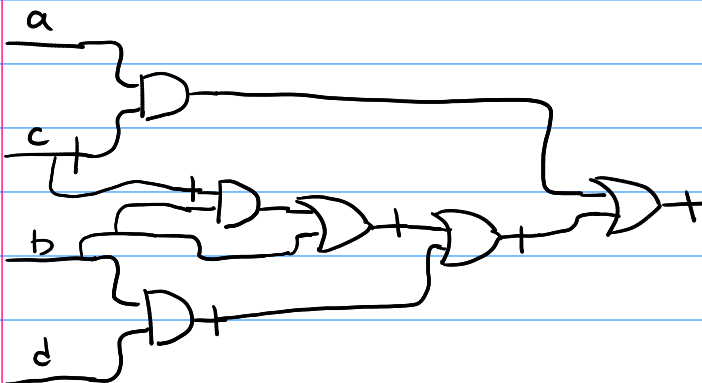
1) $\overline{a(b+c)}(a+b)$



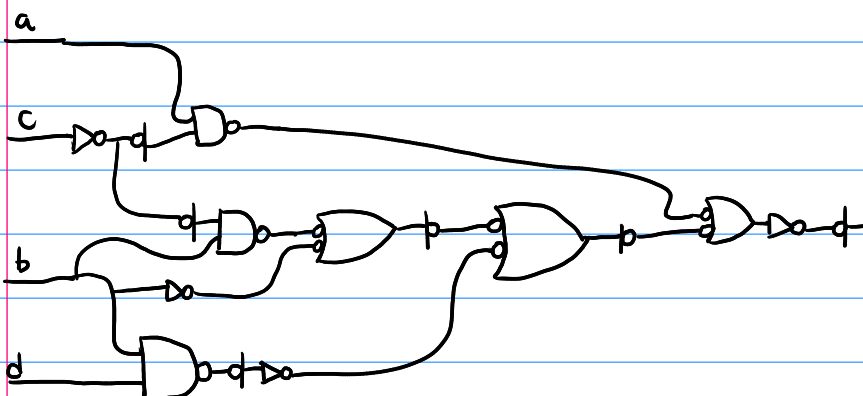
4 NOR
gates



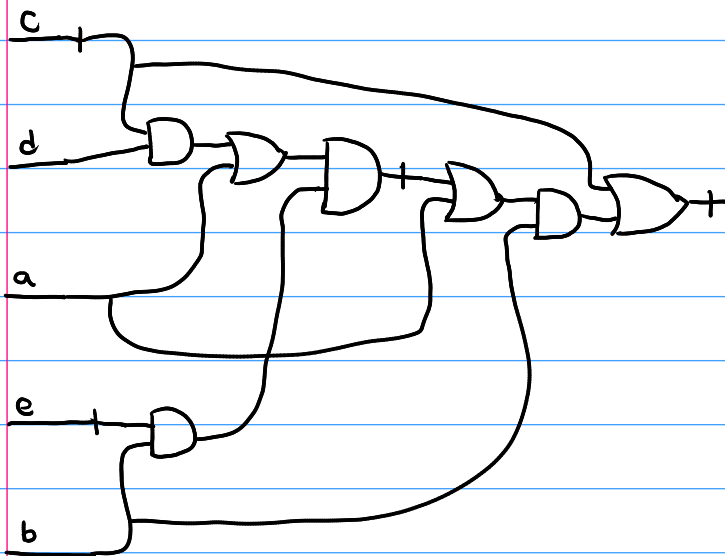
2) (2) $F = \overline{a\overline{c}} + \overline{b\overline{d}} + \overline{b} + \overline{b\overline{c}}$ using NAND2 and INV



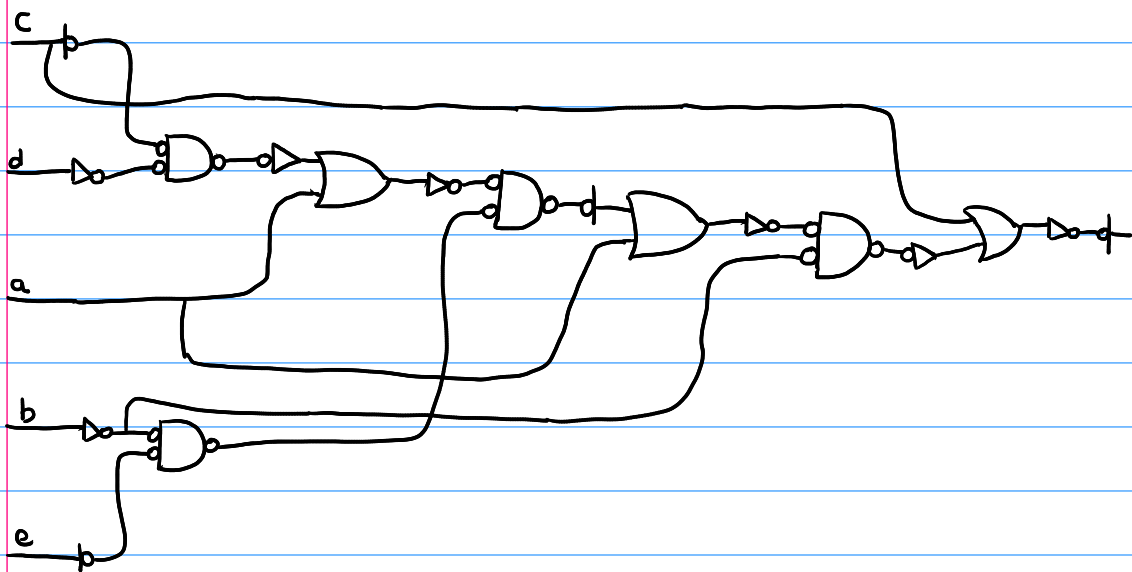
6 NAND
gates



(3) $F = \overline{\bar{c} + b(a + \overline{\bar{e}b(a + \bar{c}d)})}$ using OR2 and INV



7 OR
gates





10. And